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Relationship Between the Mobility of Aggregates and Fluid Penetration Depth Across a Range of Fractal Dimensions Using Stokesian Dynamics

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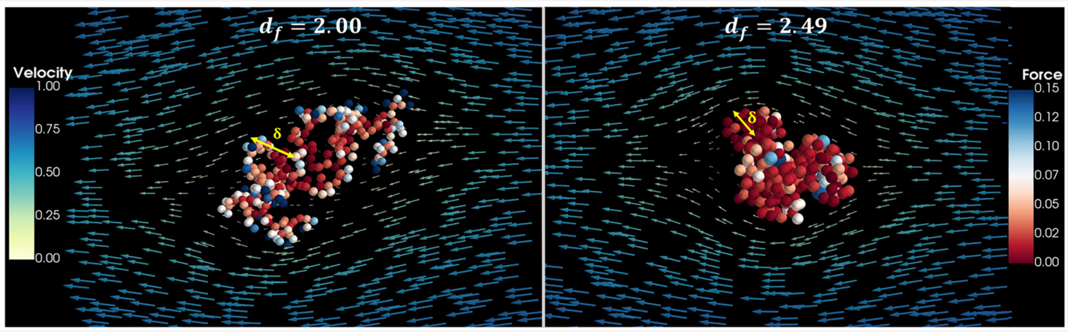

ABSTRACT: The hydrodynamic behavior of fractal aggregates plays an important role in various applications in industry and the environment, and has been a topic of interest over the past several decades. Despite this, crucial aspects such as the relationship of the mobility radius, R_m, with respect to the fractal dimension, d_f, and the fluid penetration depth, δ, have largely remained unexplored. Herein, we examine these aspects across a wide range of d_f's through a Stokesian dynamics approach. It takes into account all orders of monomer−monomer interactions to construct the resistance matrix for the entire cluster, which is assumed to be rigid. Statistical fractals created using algorithms such as diffusion limited aggregation (DLA), cluster−cluster aggregation (CCA), tunable Monte Carlo algorithm, and a deterministic Vicsek fractal, with d_f varying from 1.76 to 3, and the number of monomers ranging from 20 to 10 240 are considered. While confirming the expected asymptotic cluster-size independence of the hydrodynamic ratio, $\beta = R_m/R_g$ (where R_g is the radius of gyration of the cluster), this study reveals a monotonically increasing trend for β with increasing d_f. The decay of the fluid velocity within the aggregate is quantified via the concept of penetration depth (δ). Analysis shows that the dimensionless penetration depth ($\delta^* = \delta/R_g$) approaches asymptotic constancy with respect to cluster size in contrast to a weak dependency of the form $\delta^* \sim (R_g/a)^{-(d_f-1)/2}$, predicted by the mean-field theory (a being the monomer radius). Furthermore, the penetration depth is found to decrease rapidly, in an exponential manner, with increasing β. This establishes a quantitative relationship between the resistance experienced by the cluster and the degree of penetration of fluid into it. The implications of these results are further discussed.

1. INTRODUCTION

Understanding the mechanisms of transport of particles suspended in a fluid is of crucial importance in aerosol systems,[1−4] geosciences,[5−7] and chemical technology. In the former case, the increasing concern of the role played by particulate matter on human health,[4] global warming,[8] and bioaerosols in the context of air borne infections including COVID-19[9] is driving a large number of studies. Similarly, whether it be the transport of contaminants in groundwater[10,11] or the constant need to improve the performance of technologies using disperse phase systems in manufacturing,[12] material processing, and catalysis,[13,14] increasingly sophisticated approaches are required for elucidating fluid−particle interactions. The occurrence of particles in the form of fractal aggregates is a special case of interest. Such situations arise in the case of aerosols released in nuclear accidents,[15,16] soot formed from combustion processes,[17] alumina clusters in molten steel,[18] the addition of nanostructures for enhancing fluid properties,[19,20] and zeolites and other high surface area materials used in catalysis.[21] The origin of fractality in aggregates arises mainly because of the irreversible coagulation of the dispersed particles due to thermal or mechanical agitations. In view of their complex structure, special attention

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is required to describe their hydrodynamic and associated properties. Specifically, there exists a need to quantify the transport, deposition on surfaces, particle−particle interaction characteristics in pipes,[22] vessels, and containment structures,[15,23] and fluid penetration characteristics in catalysts. Also, in the context of aerosols, the effects of the nonspherical nature are often expressed as shape factors, which influence the assessment of internal exposure due to the inhalation of radioactive particles.[24,25]

The experimental investigation of Forrest and Witten[26] using metal vapor aerosols first established the statistical nature of fractality of the aggregates. Since then, the concept of fractal geometry has been vigorously pursued by understanding both the mechanism of aggregation and the hydrodynamic behavior of the aggregates.[27,28] Two important parameters characterize these fractals: radius of gyration (R_g) and fractal dimension (d_f). Fractal aggregates always follow the scaling relationship given by Forest and Witten[26] and Nagel:[29]

$$N = k_f\left(\frac{R_g}{a}\right)^{d_f} \tag{1}$$

where N is the number of monomeric particles of radius a and k_f is a prefactor. The radius of gyration is obtained using

$$R_g = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{r}_i - \boldsymbol{r}_{cm})^2}, \quad N > 1 \tag{2}$$

where $\boldsymbol{r}_{cm} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{r}_i$ is the coordinate of the center of mass and r_i is the position vector of the i^{th} monomer of the aggregate.

The mobility radius (R_m) is defined as the radius of an equivalent sphere experiencing the same drag force (f) as the aggregate for a specified fluid velocity magnitude (u^∞). Accordingly, it is given by

$$R_m = \frac{f}{6\pi\mu u^\infty} \tag{3}$$

where μ is the fluid viscosity. Sometimes, this relationship is also expressed via the dynamic shape factor.[30] To extract the mobility radius, the study of the influence of shape and structure on the dynamics of the aggregates is a matter of fundamental importance.

The main purpose of this paper is to understand the challenging aspects of the mobility of aggregates in relation to their sizes, fractal dimensions, and degree of fluid penetration. Historically, the Kirkwood−Riseman theory and porous sphere models have been used for understanding the hydrodynamic properties of fractal aggregates.[31] The principle of the Kirkwood−Riseman[32] theory is that the velocity of the individual particles in a cluster is affected by the presence of other particles in the same cluster, collectively causing disturbances in the velocity of the fluid both within and outside of the cluster. Another approach is to consider the fractal aggregate to be a porous sphere in the low Reynolds number region. Flow within the cluster is governed by either Darcy's law or the Brinkman equation.[31] Darcy's equation is given by $-\frac{\mu}{\kappa}u = \nabla p$, where κ is permeability, which is a function of the volume fraction of the particles, and u and p are fluid velocity and pressure, respectively.[33] Permeability is a critical parameter for finding the drag force acting on an aggregate using correlations. Another alternative is to use Brinkman's equation given by $\mu\nabla^2 u - \frac{\mu}{\kappa}u = \nabla p$. Brinkman's modification provides an equation that interpolates between the Stokes equation and Darcy's law. Brinkman's equation is better suited to handle homogeneous permeable structures as shown by Beavers and Joseph;[34] fractal aggregates have a structure with varied permeability.[35] Models used by Veerapaneni and Wiesner[35] considered permeability to be proportional to the radial distance (increasing radially) from the center of mass to the outer radius. Permeability in the models of Kim and Yuan[36] increases quadratically from the center, $\kappa \propto r^2$.

In 1979, using the Kirkwood−Riseman approximation, De Gennes[38] showed that R_m should be proportional to R_g. Later, Wiltzius[37] provided an experimental demonstration of the proportionality based on static and dynamic light scattering techniques for large aggregates. From a practical perspective, the ratio $\frac{R_m}{R_g}$, referred to as the hydrodynamic ratio, is the quantity of crucial interest. It is apparent from the linear correlations between the two that the hydrodynamic ratio obtained by the application of the Kirkwood−Riseman theory is a function of fractal dimension only,[39] regardless of the cluster mass. However, this is an asymptotic result valid for large clusters, and the work of Chen et al.[40] with diffusion limited aggregation (DLA) and bond percolation aggregates points at the dependence of the hydrodynamic ratio on N. Specifically, $\frac{R_m}{R_g}$ increases for DLA as N increases and then reaches an asymptotic value. However, there appears to be no universality of the trend as evidenced by the study of Warren[41] using porous sphere model simulations, which showed an increase of $\frac{R_m}{R_g}$ with N for DLA for which the density is the highest in the center, whereas a decreasing trend leading to quick saturation was observed for clusters formed via Diffusion Limited Cluster-Cluster Aggregation (DLCCA). In other words, there appears to be certain nonuniversality with respect to the transient behavior of the hydrodynamic ratio, being specifically dependent upon the type of the aggregate.

Van Saarloos[39] further developed the transient response model beyond the Kirkwood−Riseman theory, thereby demonstrating that the hydrodynamic ratio will be dependent not only on the fractal dimension but also on the size of the aggregate and the degree of screening. Van Saarloos[39] expressed the inverse permeability in terms of the cluster's outer radius (or the cutoff radius, R_c), which is linearly related to R_g and scales with N through the same fractal exponent. The hydrodynamic ratio is then obtained with the knowledge of fractal dimension, d_f, and permeability, κ.

As suggested by Gmachowski,[42] the hydrodynamic ratio increases as the fractal dimension increases and reaches a maximum value when $d_f = 3$. This is explained by the fact that the internal permeability within the aggregate increases with increasing fractal dimension. Similar conclusions were reached in the works of Fellay et al.,[43] Chopard et al.,[44] and Harshe et al.[45]

In an important study, Nguyen[31] observed different hydrodynamic ratios for two different clusters with the same fractal dimension. This was attributed to the fact that the two aggregates had different coordination numbers within their regions. The coordination number is the average number of monomers connected to a particle in an aggregate. Vanni[46] made an observation of the effect of the coordination number

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on the shielding of fluid flow. The resistance of each sphere is said to decrease as more and more spheres are in contact.[46]

The variations in the findings of previous studies could largely be attributed to the use of different approaches by different authors. This requires a need to examine the hydrodynamic behavior across a wide range of d_f, N, and fractal types using a single methodological framework. Furthermore, interdependencies between the observed resistances, fractal dimensions, and the degree of fluid penetration have remained largely unexplored. In this paper, we focus on these aspects by conducting hydrodynamic simulations through a Stokesian dynamics approach for aggregate particles in the continuum regime. The extension of the approach to noncontinuum regimes needs to be explored separately in the future. The desired fractal aggregates are generated using different algorithms as described in Section 2. The relationships between various structural and hydrodynamic quantities, including fluid penetration σ, are discussed in Section 3. The implications of the findings are discussed in the concluding Section 4.

2. METHODOLOGY

Three-dimensional aggregates composed of N primary particles are generated and then simulated for their hydrodynamic properties. Both mathematical and statistical fractal clusters were used in the study. Further, the statistical fractals were generated using both random-walk as well as tunable Monte Carlo methods. These are described below.

2.1. Generation of Fractal Aggregates. In aerosols and colloids, different mechanisms govern the process of aggregate formation depending on the concentration of monomers. Here, we use three algorithms to simulate the generation of statistical fractals with different fractal dimensions.

The first algorithm is that of irreversible particle−cluster aggregation, which is prevalent in the dilute limit, i.e., diffusion limited aggregation (DLA).[47] Here, monomers diffuse one-by-one to a stationary cluster and become incorporated in it, resulting in the growth of the structure with a fractal dimension (d_f) of 2.5. Using an in-house code for accelerated off-lattice aggregation with varying step sizes, three instances each of clusters with as many as 10 000 monomer spheres were created. The firing radius was maintained to be greater than 20 times the maximum cluster radius to prevent the formation of asymmetric side arms.

The second mechanism of aggregate formation is relevant when the monomer concentration is higher, thus facilitating the formation of multiple clusters, which in turn collide and get aggregated with each other forming larger clusters. The cluster−cluster aggregation (CCA) model describes the formation of porous structures like silica gels and allophane aggregates.[48] Most combustion generated aggregates are also said to fall under this category.[49] These clusters are generated using the application created by Li and Xiong,[50] which is an on-lattice CCA algorithm. Using the initial concentration (C) of 0.001 monomers per lattice site, DLCCA clusters of fractal dimension, d_f = 1.84, were created. Similarly, CCA clusters of fractal dimensions 2.21 and 2.12 were created with concentration of 0.01 monomers per lattice site using two different diffusion exponents, viz., $\gamma = 0$ and -0.5, respectively. The lattice size, L, was varied from 22 to 215 to create clusters with number of monomers ($N = CL^3$) between 10^2 and 10^4. The sticking probability, σ, was maintained as unity.

The third cluster generation method used here is based on the algorithm proposed by Filippov et al.[51] to generate clusters “tuned” to a particular fractal dimension and prefactor. For future purposes, we refer to these as tunable clusters. The “*aggregate_gen*” package was downloaded from GitHub.[52] The algorithm starts with precursor spherules of desired fractal dimensions comprising 4 to 10 particles that undergo subsequent attachment while maintaining the desired fractal dimension and prefactor. When the number of the levels of attachment were varied between 2 and 10, clusters of sizes ranging from 20 to 10 240, fractal dimensions ranging between 1.76 and 2.49, and unit prefactor were created.

In addition to the above three types of statistical fractals, Vicsek fractals with N = 7, 49, 343, and 2401 with an exact fractal dimension of $d_f = {}^{\ln 7}/_{\ln 3}$ and completely filled spherical aggregates with monomers arranged in a simple cubic lattice were used for benchmarking. Figure 1 shows a summary of the range of fractal dimensions studied. The detailed conditions used for the generation of the above three categories of fractal aggregates are summarized in the Supporting Information.

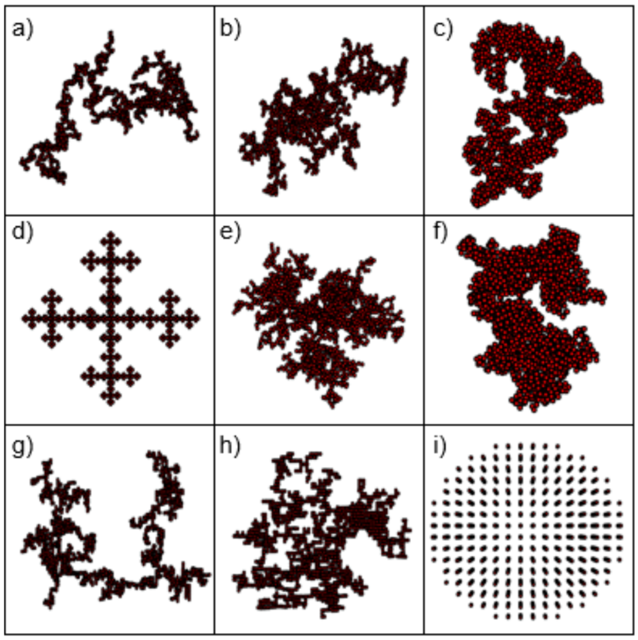

Figure 1. Typical aggregates generated using different models such as tunable Monte Carlo clusters of fractal dimensions (a) 1.76, (b) 2.00, (c) 2.25, and (f) 2.49 and N = 2048; (d) Vicsek with N = 2401; (e) DLA with N = 2000; (g) DLCCA with d_f = 1.89 and N = 2048; (h) CCA with d_f = 2.21 and N = 2160; (i) simple cubic unit with N = 2176.

The interparticle distance is maintained at 2.01 giving us particles of unit radius with an interparticle spacing of 0.01 radial units. The radius of gyration in the tunable CCA algorithm was calculated as

$$R_g^{\,2} = a^2 + \frac{1}{N}\sum_{i=1}^{N} |(\boldsymbol{r}_i - \boldsymbol{r}_{cm})|^2 \tag{4}$$

where $\boldsymbol{r}_i$ is the position vector of the i^{th} particle, $\boldsymbol{r}_{cm}$ is the coordinate of the center of mass of the aggregate. The monomer radius, a, is maintained as unity throughout this study. The outer radius of the aggregates, R_c, is calculated as half of the mean of square of the span in all three Cartesian directions $R_c = \sqrt{\frac{R_x^{\,2} + R_y^{\,2} + R_z^{\,2}}{3}}$, where R_i is half the axis length in the i direction.

2.2. Hydrodynamic Simulations. An F−T (force−torque) version of the Stokesian dynamics simulation method, first proposed by Brady and Bossis,[53] has been used for calculating the hydrodynamic resistance experienced by the fractal aggregates under creeping flow conditions. We use a modified version of Townsend’s implementation of the Stokesian dynamics code in Python.[54] The algorithm was tested for linear chains of 5, 7, 9, and 25 spheres with varying interparticle distances of 2.005, 2.2, and 2.6 and for cases of spheres at vertices of a triangle or square, and the calculated drag coefficients and trajectories were found to be in agreement with those reported by Durlofsky and Brady.[55] Further, in order to maintain the structural integrity of the clusters, Townsend’s code was modified to

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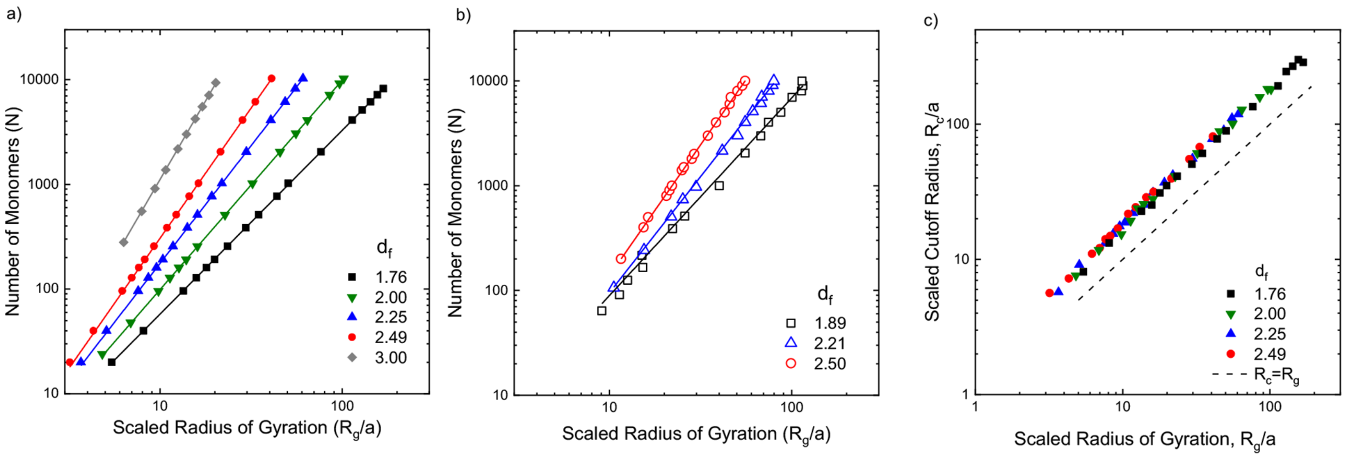

Figure 2. Illustration of power-law scaling of the number of monomers (N) in the aggregate with respect to the scaled radius of gyration (R_g/a) in the range $N = 20$ to $N = 10\,240$ for (a) tunable Monte Carlo aggregates (■, $d_f = 1.76$; green ▼, $d_f = 2$; blue ▲, $d_f = 2.25$; red ●, $d_f = 2.49$) and simple cubic lattice (gray ◆, $d_f = 3.00$) using one realization per cluster size; (b) random-walk fractals (□, $d_f = 1.89$; blue △, $d_f = 2.21$; red ○, $d_f = 2.50$) with up to ten realizations for small clusters for each $N < 200$, at least 3 realizations for large clusters for every $N > 2000$, and an intermediate number of realizations for intermediate cluster sizes. Lines are power-law fits. (c) Illustration of the scaling of the cutoff radius (R_c/a) for all the clusters with respect to the radius of gyration (R_g/a) for tunable Monte Carlo clusters along with a dashed line representing unit slope (— — —) and ■, $d_f = 1.76$; green ▼, $d_f = 2$; blue ▲, $d_f = 2.25$; red ●, $d_f = 2.49$.

incorporate the rigid body formulation used by Swan et al.[56] Similar approaches have been used by Harshe et al.[45] for spherical particles and by Kutteh[57] for arbitrary shaped rigid bodies. Thus, each particle within the aggregate translates with the same velocity, and the relative interparticle configuration remains unchanged. For an aggregate composed of N particles, each of which is subject to an external force and/or torque, the aggregate velocity is calculated as

$$\mathbf{U} = -(\mathbf{\Sigma}\cdot\mathbf{R}_{\mathrm{FU}}\cdot\mathbf{\Sigma})^{-1}\cdot\mathbf{\Sigma}\cdot\mathbf{F}^{\mathrm{P}} \tag{5}$$

where $\mathbf{U} = [U_x, U_y, U_y, \Omega_x, \Omega_y, \Omega_z]$ is a 6×1 vector representing translational (U) and rotational (Ω) velocity components and $\boldsymbol{F}^{\mathrm{P}} = [F_x^1, F_y^1, F_y^1, T_x^1, T_y^1, T_z^1, ...]$ is an $6N \times 1$ vector representing the force and torque acting on the each of the N particles. The $6N \times 6N$ configuration dependent resistance matrix $\mathbf{R}_{\mathrm{FU}} = M^{\infty-1}$ is obtained via inversion of the mobility matrix M^∞. M^∞ is assembled by pairwise addition of first few moments from the multipole expansion of the force density on the surface of the particles in a series of moments to yield a far-field approximation, the inversion of which is equivalent to an infinite series of reflected resistance interactions and takes into account many body interactions.[58] $\mathbf{\Sigma}$ is a $6N \times 6$ tensor responsible for summing the forces and torques acting on the individual spheres to yield the net force and torque acting on the aggregate, and is used to obtain rigid body motion of the aggregate. All the quantities are rendered dimensionless using a length scale equal to the monomer particle radius, a, time scale of a/V and force scale of $6\pi\mu Va$, where V is a characteristic velocity and μ is the fluid viscosity. Thus, a single particle experiencing a unit force in a given direction translates with unit velocity in the same direction. As input to the Stokesian dynamics calculation, the cluster configurations obtained in Section 2.1 are scaled such that all aggregates are composed of monomeric particles of unit radius and the gap between neighboring particles is fixed at 0.01. Since the simulations are confined to rigid structures only, there will not be any relative motion between the neighboring particles. For a pair of such rigidly held particles, the nearfield lubrication contribution to hydrodynamic drag would be less than $O(0.01)$[59] and therefore the nearfield lubrication interactions have been excluded, as also previously justified by Bossis et al.,[60] Harshe et al.,[45] and Swan et al.[61]

For the calculation of hydrodynamic resistance offered by an aggregate, a unit force is applied to each constituent sphere. The ratio of the total force of N units exerted on the aggregate to the resulting aggregate velocity gives the translational mobility radius in that particular direction. The overall mobility radius is calculated by averaging the derived mobility radius in three Cartesian directions. The effect of any small rotational velocity induced due to asymmetry of the clusters is accounted for by the $\Omega-T$ coupling.

Given the force $\boldsymbol{F}^\alpha$ acting on α^{th} particle, the fluid velocity $\boldsymbol{u}$ at any point $\boldsymbol{x}$ in the fluid is calculated as a superposition of multipole expansions around each particle center, $\boldsymbol{x}^\alpha$:

$$u_i(x) = u_i^\infty(x) - \frac{1}{8\pi\mu}\sum_\alpha \left[-\left\{1 + \frac{1}{6}a^2\nabla^2\right\}\boldsymbol{J}_{ij}(x - x^\alpha)F_j^\alpha + \frac{1}{2}\varepsilon_{jkl}\nabla_k\boldsymbol{J}_{il}(x - x^\alpha)T_j^\alpha\right] \tag{6}$$

where $\boldsymbol{J}$ is the Oseen's tensor. With the aggregate being kept rigid in a stream of fluid translating with uniform velocity of unit magnitude $\mathbf{U}^\infty$, the hydrodynamic forces and torques acting on the individual particle(s) are calculated as $\mathbf{F} = \mathbf{R}\cdot(\mathbf{U} - \mathbf{U}^\infty)$ and used in eq 6 to yield the fluid velocities in the interior regions of the cluster, taking into account all orders of the perturbations by the particles. The fluid velocity is estimated on a fine Cartesian grid of up to $100 \times 100 \times 100$ points. We use the scalar quantity, namely, the magnitude of velocity at a point, as a measure of fluid penetration at that point. The velocity magnitude versus radial distance r obtained by radial binning for each cluster is fitted to obtain a penetration depth. This exercise is performed for three Cartesian directions of fluid flow to obtain three penetration depths that are then averaged. Further averaging is performed over four independent cluster realizations for $d_f = 1.76$ and two realizations for $d_f = 2.49$ for every N and appropriate number of realizations for intermediate fractal dimensions.

3. RESULTS AND DISCUSSION

3.1. Cluster Geometry and Structural Parameters. The fractal dimensions (d_f) and prefactors (k_f) are obtained by linear regression to the log–log plots of N versus R_g/a as shown for tunable Monte Carlo (TMC) clusters in Figure 2a and for random-walk aggregates in Figure 2b. Aggregates with $N > 20$ have only been considered to exclude the effects of lower cutoff transients in smaller clusters. TMC clusters, being regulated in respect of their d_f's, have lower error in the fitted quantities despite using just one realization per cluster size N. This is because the rules for generating the clusters are constrained to satisfy the scaling relations at all stages. To

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ensure sufficient sampling for random-walk aggregates (DLA, CCA, and DLCCA), up to three different clusters for a given N, for $N > 2000$, and up to 10 clusters per given N, for $N < 200$, were used. For DLAs, the theoretical exact fractal dimension is well-known to be 2.5 in 3D. The fractal dimension of DLA clusters generated in this study is 2.498 ± 0.056, which is similar to those achieved by other investigators.[40,44] In the case of DLCCA, the maximum standard error in the estimates of d_f was 2.6%. Table 1 shows a summary of the results for the three types of fractals generated across a fractal dimension ranging from 1.76 to 2.50. In the case of DLCCA, the fractal dimension achieved depends on the initial monomer volume fraction (C) as well as the diffusion coefficient exponent (γ).

Table 1. Fractal Dimension (d_f) and Prefactor (k_f) Observed for Various Fractals[a]

fractal type			fractal dimension (d_f)	prefactor (k_f)
Tunable Aggregates				
tunable Monte Carlo cluster−cluster aggregate			1.76 ± 0.003	1.016 ± 0.016
			2.00 ± 0.000	0.992 ± 0.001
			2.25 ± 0.000	0.996 ± 0.002
			2.49 ± 0.001	1.002 ± 0.001
simple cubic lattice			3.00 ± 0.000	1.125 ± 0.000
Random-Walk Aggregates				
DLCCA	$C = 10^{-3}$	$\gamma = 0$	1.890 ± 0.060	1.164 ± 0.024
CCA	$C = 10^{-2}$	$\gamma = 0$	2.214 ± 0.023	0.584 ± 0.008
	$C = 10^{-2}$	$\gamma = -0.5$	2.118 ± 0.093	0.827 ± 0.232
DLA			2.498 ± 0.056	0.441 ± 0.097
Mathematical Fractal				
Vicsek			1.750 ± 0.010	2.275 ± 0.072

[a] C is the initial monomer concentration, and γ is the cluster diffusion exponent. Confidence intervals for d_f and k_f are obtained from the fitting algorithm.

Other measures of cluster size such as the outer radius (R_c) (half the average cluster span) was also attempted for scaling laws. R_c is an indicator of the span of the fractals, which is generally larger than R_g. However, while the log−log fits between R_c and R_g (Figure 2c) yielded a consistent linear relation between the two (refer to Figure 2c), as seen from the fact that it is parallel to the unit slope line in the log−log scale, the errors obtained when fitting R_c with respect to N were larger (not shown in the figure), thereby rendering R_c as a relatively less reliable metric for system size. Hence, we adopt the radius of gyration (R_g) as the measure of size throughout this paper to elucidate the various scaling laws.

3.2. Mobility Radius versus Cluster Size. The relationship between the mobility radius, R_m, as defined in (3), and the radius of gyration, R_g, as defined in (1) and (2), is of crucial importance for further analysis. In some applications where cluster mass (N) is the measured quantity, it is desirable to establish the dependency with respect to N and accordingly a correlation is sought in a form similar to that presented in Sorensen:[62]

$$\frac{R_m}{a} = k_m N^x \tag{7}$$

Figure 3 illustrates the scaling behavior of the mobility radius, R_m, with respect to N for TMC clusters. The fitting parameters, exponent, x, and prefactor, k_m, are presented in Table 2. For mobility radius estimates, one realization is used for TMC aggregates, and between three and ten realizations are used for random-walk aggregates (DLA, CCA, and DLCCA). In addition, orientational averaging was performed over three Cartesian directions. It was also confirmed that further averaging would not result in a significant change in estimation of the hydrodynamic radius (less than 1% change for TMC and DLA and less than 4% for DLCCA, refer to Table S2).

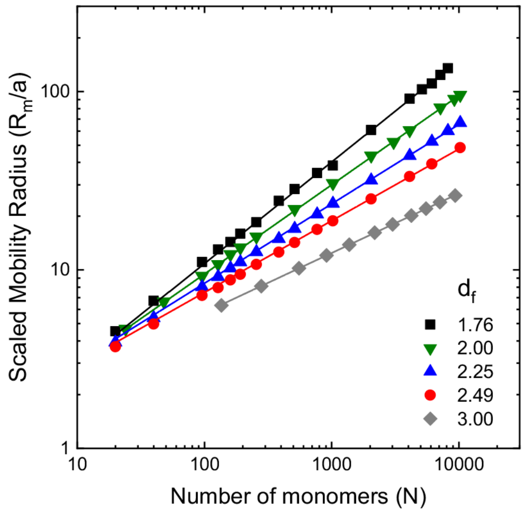

Figure 3. Mobility radius observed for tunable Monte Carlo fractals of different fractal dimensions from $N = 20$ to $N = 10\,240$. Lines are power-law fits. Each data point is obtained by averaging over three Cartesian directions of flow for one realization.

By focusing on two fractal situations, namely, diffusion limited cluster-cluster aggregates and reaction limited cluster-cluster aggregates, Sorensen showed that in the continuum limit x is close to $1/d_f$. When this is combined with the relation $R_g \sim N^{1/d_f}$, it implies that $R_m \sim R_g$. This is in conformity with the prediction of De Gennes[38] that the hydrodynamic radius, R_m, is linear with the radius of gyration (R_g) based on mean-field arguments and Wiltzius's[37] findings from light scattering experiments. From Table 2, it is seen that the values of the exponent x obtained in the present study are also close to $1/d_f$, thereby extending the validity of the linear relation between R_m and R_g over a wider class of aggregates covering a broader range of fractal dimensions. However, the agreement between x and $1/d_f$ is an asymptotic result and might be attained very slowly. This leads to $1/d_f$ and x differing perceptibly (within 7%) for DLA and DLCCA. We find that the agreement increases with increasing cluster size (refer to Table S3). Seto et al.[63] report a similar behavior for Vicsek fractals with $1/d_f = 0.579$ and $x = 0.527$ for $N = 7$ to 343. In the same range, we obtain $1/d_f = 0.579$ and $x = 0.525$.

Interestingly, the prefactor k_m is seen to depend on fractal type and not just the fractal dimension (d_f), as is demonstrated by the difference seen in the cases of $d_f = 2.49$ (TMC) and $d_f = 2.50$ (DLA). This shows a fundamental nonuniqueness of the prefactors of dynamic properties with respect to fractal dimensions, which is expected due to the fact that d_f alone does not uniquely capture all aspects of the aggregates. For practical purposes, one may associate unique prefactors by averaging over fractal types. It may be noted that, for the special case of DLCCA with $d_f = 1.89$, we obtain the equation $R_m/a = 0.992\ N^{0.52}$. This is quite close to the correlations of

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Table 2. Values of the Exponent, x, d_f^{-1}, and Prefactor, k_m, Obtained from Fits to Eq 6 in Relation to the Mobility Radius, R_m, and the Number of Primary Particles, N, in the Cluster

fractal type			d_f	$1/d_f$	x	k_m
			Tunable Aggregates			
tunable Monte Carlo cluster−cluster aggregate			1.76 ± 0.003	0.570	0.568 ± 0.006	0.796 ± 0.039
			2.00 ± 0.000	0.500	0.496 ± 0.002	0.984 ± 0.021
			2.25 ± 0.000	0.445	0.454 ± 0.001	1.006 ± 0.010
			2.49 ± 0.001	0.402	0.410 ± 0.001	1.098 ± 0.006
simple cubic lattice			3.00 ± 0.000	0.334	0.334 ± 0.000	1.238 ± 0.001
			Random-Walk Aggregates			
DLCCA	$C = 10^{-3}$	$\gamma = 0$	1.890 ± 0.060	0.529	0.515 ± 0.009	0.992 ± 0.078
CCA	$C = 10^{-2}$	$\gamma = 0$	2.214 ± 0.023	0.452	0.452 ± 0.014	1.381 ± 0.163
DLA			2.498 ± 0.056	0.400	0.427 ± 0.004	1.366 ± 0.044
			Mathematical Fractal			
Vicsek			1.750 ± 0.010	0.571	0.530 ± 0.010	0.820 ± 0.030

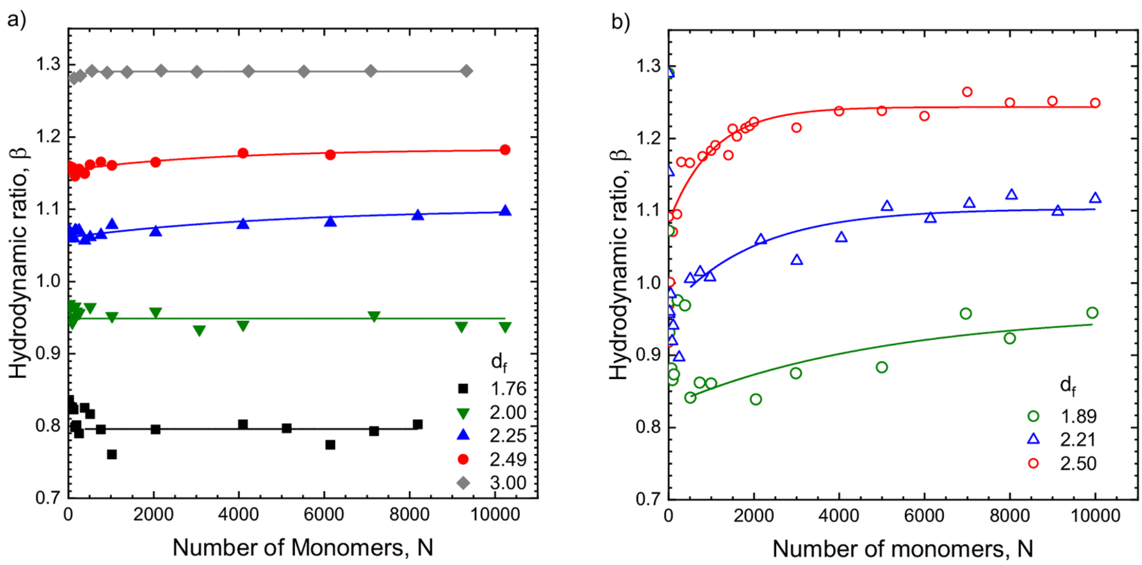

Figure 4. Variation of the hydrodynamic ratio, β, with the aggregate mass, N, for (a) tunable Monte Carlo clusters and simple cubic lattice, and (b) random-walk aggregates. The continuous lines represent the fitted graphs using a crossover function from eq 7. For TMC clusters in (a), each data point is obtained by averaging over three Cartesian directions of flow for one realization. Between three and ten realizations are used for random-walk aggregates in (b).

Dastanpour and Rogak[64] and Sorensen[62] for the cases of DLCCA with $d_f = 1.78$.

3.3. Hydrodynamic Ratio, β. Given the underlying linearity between R_m and R_g, one may define a quantity of practical interest, namely, the hydrodynamic ratio ($\beta = R_m/R_g$), which is a prefactor determined by the slope of the linear plot between R_m versus R_g. Previous studies by Fellay et al.,[43] Chen,[40] and Gmachowski[42] were confined to small N while the study by Nguyen[31] was conducted for large N using the Lattice Boltzmann method (LBM). In this paper, we extend the investigations across a large N and wider range of d_f using a common formalism of Stokesian Dynamics. The quantity β may be directly evaluated by correlating R_m and R_g for each cluster type across entire range of N. The transient effects due to finite cluster are handled by choosing a crossover function with respect to N from which an asymptotic value may be evaluated by fitting. The asymptotic value of the hydrodynamic ratio denoted by β_∞ is estimated by fitting the data to a saturating exponential function of the form:

$$\beta = \beta_\infty - Ae^{-N/N_0} \tag{8}$$

where N_0 is the cluster size beyond which the constancy is reached and A is a constant. The detailed N dependence of β for N up to 10 000 for the various aggregate types is shown in Figure 4 along with the fitted plots. Previously, Chen et al.[40] have estimated β for DLA clusters up to $N = 900$ using the Rotne−Prager tensor formalism. Seto et al.[63] estimated hydrodynamic radii using Stokesian dynamics up to $N = 1000$ for DLA and up to $N = 400$ for Vicsek fractals. With respect to DLA, special mention should be made of the results obtained by Seto et al.[63] and Chen et al.[40] for the dependence of the hydrodynamic ratio with respect to N. The slow saturation trend observed in the present study is also seen in these studies. The transient nature of β with N obtained in this work almost superimposes on those reported by Seto et al.[63] and Chen et al.[40] (refer to Figure S1).

As may be seen, β possesses a weak N dependence for TMC clusters. β_∞ is the quantity of practical interest. Figure 5 shows a dependence of asymptotic hydrodynamic ratio, β_∞, on the fractal dimension, d_f, from this study as well as a comparison with previous works.

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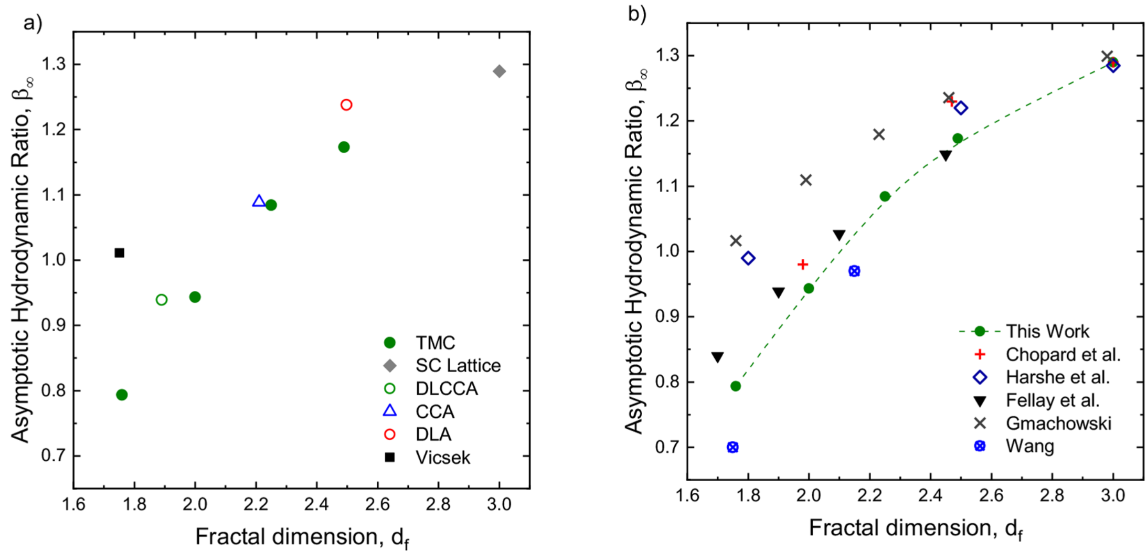

Figure 5. (a) Dependence of asymptotic hydrodynamic ratio, β_∞, versus fractal dimension, d_f, for different fractals of this study. (b) Comparison of the present estimates for TMC clusters with previous works on CCA estimated by Chopard et al.[44] using the Lattice Boltzmann method, Harshe et al.[45] using the Stokesian dynamics method, Gmachowski[42] using the porous sphere model, and experimentally measured by Wang and Sorensen.[66]

There exists a weak dependence of β_∞ with respect to fractal types as noted previously and indicated by various studies.[39,42] Notwithstanding this, the general trend is that fractals with higher fractal dimension have a larger hydrodynamic ratio as compared to the fractals with lower fractal dimension. The maximum value of $\beta_\infty = R_m/R_g$ was observed to be 1.29 for a simple cubic unit having a fractal dimension of 3, as expected for a solid sphere and is the maximum limiting value of R_m/R_g in the literature.[39,43,45]

On the other hand, for fractals with $d_f < 2$, R_m/R_g turns out to be less than unity. This is a reflection of the enhanced transparency of sparser objects for fluid flow into their interior regions and the consequent reduction in their resistance (see the next section).

A comparison of the β values obtained for the tunable Monte Carlo clusters with the literature data shows that our results are consistent with general observations by Fellay et al.,[43] Chopard et. al,[44] Gmachowski,[42] and Harshe et al.[45] While Chopard et al.[44] employed LBM for $100 \leq N \leq 20\,000$, Harshe et al.[45] and Fellay et al.[43] used Stokesian dynamics for small N. Gmachowski[42] uses the Brinkman equation on cluster−cluster aggregates obtaining an exceptional value of β exceeding unity for $d_f = 1.75$, which is not the case with any other method. As fractal dimension increases (up to $d_f = 2.45$), the difference between the results of Gmachowski[42] and those of the present study as well other studies narrows down considerably.

While studying the hydrodynamics of colloidal silica aggregates with $d_f = 2.1$, Wiltzius[37] obtained β of 0.72 experimentally. Pusey and Rarity[65] later corrected this to 0.98. At $d_f = 2.15$ with 0.95, Wang and Sorensen[66] obtained a β value that is similar to the current investigation. However, their β of 0.7 at $d_f = 1.75$ is low in comparison to other results of a similar fractal dimension.

Dastanpour and Rogak[64] obtained a hydrodynamic ratio of 0.89 ± 0.02 for DLCCA ($d_f = 1.78$) with $64 \leq N \leq 512$, where the values obtained were consistent with the work of Chen et al.[40] It is comparable to our 0.94 ± 0.01 observed for DLCCA ($d_f = 1.89$). Thus, different methodologies yield the same values for compact aggregates while β for an arbitrary d_f is not exactly the same but within close range. Although the hydrodynamic ratio for Vicsek fractals appears to be an exception to the trend with respect to d_f, it is consistent with past studies wherein a similar exceptional behavior was observed by Filippov et al.[51] and Seto et al.[63] (refer to Figure S1).

From Figure 5, a general observation can be made that the hydrodynamic factor (β_∞) increases with the fractal dimension. This fact implies that for an aggregate of a given size hydrodynamic resistance increases with the fractal dimension. This calls for further investigation into the underlying mechanism of fluid flow within the cluster.

3.4. Fluid Velocity and Penetration Depth. The basic distinguishing feature of hydrodynamics of the aggregates from that of spherical objects is the question of fluid penetration into the interior regions of the objects. For given aggregate, the higher the fluid penetration, the lower will be the resistance and, correspondingly, the higher will be the mobility radius. There must be some correlation between the mobility prefactor, β_∞, fractal dimension, d_f, and the extent of fluid penetration. The concept of fluid penetration has been investigated in the past. Debye and Bueche[67] used a "shielding length" to study the exponential decay of velocity for the case of polymer beads. Hwang et al.[68] obtained "penetration depth" for measuring the distance that the fluid can penetrate into a porous medium with a similar exponential decay of velocity. Kim and Yuan[36] obtained fluid velocity indirectly by determining the velocity of a tracer particle of size $10^{-3}a$ flowing through the aggregate without overlapping. This was compared with velocities obtained using Davies'[69] permeability equation where it matched with the Davies' value at the outer region where fluid penetration is maximum (refer to Figure S2).

We utilize the concept of "penetration depth" as a quantitative measure of fluid penetration into the aggregate. Penetration depth may be defined as the distance from the cluster surface toward its interior at which the fluid velocity magnitude falls by a factor $1/e$ and is estimated from its decay profile within the aggregate. It is to be expected that, as one moves away far from the aggregate along the radial coordinate r, the velocity disturbances decay as $\sim 1/r$.

From the Brinkman extension of the mean-field theory,[39] the collective effect of the monomers within the aggregate is to introduce velocity screening, which implies an exponential form for the decay of its profile as one moves toward the center from the surface. Formally, it may be written as $u^*(r^*) = u_s \exp\left(-\frac{R_c^* - r^*}{\delta^*}\right)$, where $u^*(r^*)$ is the dimensionless magnitude of velocity (u/u^∞) of the fluid at a dimensionless radial distance, $r^* = r/R_g$, within the cluster, u_s is a parameter representing the magnitude of velocity at the aggregate surface $\left(\boldsymbol{r}^* = \boldsymbol{R}_c^* = \frac{\boldsymbol{R}_c}{\boldsymbol{R}_g}\right)$, and $\delta^* = \frac{\delta}{R_g}$ (δ being the penetration depth). We can absorb $e^{-R_c^*/\delta^*}$ into the prefactor and rewrite $u = u_i \exp\left(\frac{r^*}{\delta^*}\right)$. The decay of fluid velocity as seen from the exterior is now transformed as an exponential increase with respect to radial distance, r, as seen from the interior core. A log−linear fit to the simulation data for u versus r^* would yield δ^*.

Figure 6 shows the variations of u^* as a function of r^* for different fractals in the log−linear scale. We observe a nearly

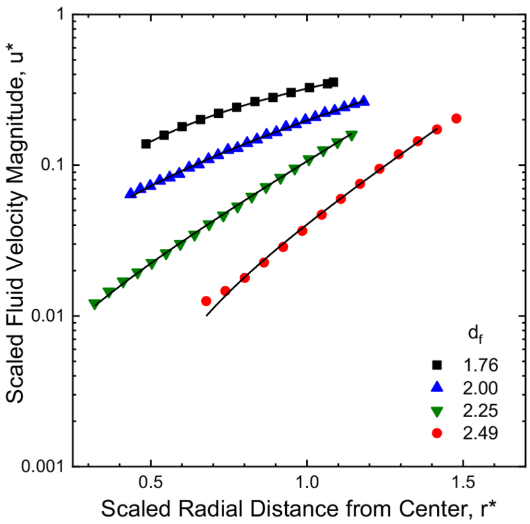

Figure 6. Fluid velocity magnitude normalized by the ambient fluid velocity as a function of nondimensional radial distance, r^*, for tunable Monte Carlo clusters with varying fractal dimensions along with the exponential fits (solid lines) using $u = u_i \exp\left(\frac{r^*}{\delta^*}\right)$, which gives the scaled penetration depth, δ^*. $N = 1024$ and ambient fluid velocity magnitude, u^*, = 1.

linear decrease, indicating an exponential fall in the magnitude of the velocity in the region carrying most of the mass (~50% to 90% of particles) located in the middle portion of the cluster ($0.5 < r^* < 1.2$). Regions near the external periphery as well as the interior regions suffer from statistical limitations and hence are not included for the exponential fits.

3.4.1. Dependence of Penetration Depth, δ^*, on Cluster Size, N. The variation of δ^* with respect to N is shown in Figure 7 for TMC clusters. The first two cases ($d_f = 1.76$ and $d_f = 2$) shows some oscillations because of the variations pertaining to the structure in different directions. Seen as a whole, across the range of d_f examined here, there is a clear indication of saturation for δ^* in the region above $N = 200$ and beyond, thereby suggesting the possibility of its cluster size independence as $N \to \infty$. This is in disagreement with the mean-field theory result, which predicts a significant decay in the same range of cluster sizes. For example, let us consider the van Sarloos mean-field expression $\delta^* \sim N^{-\left(\frac{d_f - 1}{2d_f}\right)}$ for the case of $d_f = 2.23$.[39] This yields a value of 0.43 for the ratio of δ^* between $N = 2000$ and $N = 100$, which is far smaller than that observed in the simulations (0.88). That is to say, δ^* varies very little as the cluster size increases. In fact, several other correlations exist in the literature, which show a slower decay of the penetration depth as compared to the mean-field theory. In particular, the correlation based on the empirical permeability expression proposed by Davies[69] and Kim and Russel[70] shows the slowest decay, almost in close agreement with our simulation observations. Considering that the functional forms used in the literature are empirical and not based on theories, the weak dependency of δ^* on N may as well be an artifact of statistical fluctuation and an asymptotic constancy of δ^* may not be ruled out. To confirm this, we performed a transient analysis by fitting the data to a generalized exponential decay of the form $\delta^* = \delta_e^* + A_o \exp\left(-\frac{N}{N_o}\right)$, where A_o and N_o are constants, which makes an allowance for a finite, nonzero δ^*. The results of these fits are shown in Figure 7 along with the simulation data. These values denoted by δ_e^* are listed in Table 3. The exponential relation serves as a transition function, from a finite size to the asymptotic limit, and allows us to obtain the asymptotic values of δ_e^* in spite of fluctuations for $d_f = 2.25$ and 2.49. For lower fractal dimensions, the data remains noisy despite an increase in the number of repeats and we evaluated the asymptotic constancy as a simple average across the data. For all practical purposes, we can consider these values as representative of the asymptotic limit of the penetration depth.

As may be seen in Table 3, there exists a monotonic inverse dependence of the penetration depth on the fractal dimension. The smaller the fractal dimension, the larger is the penetration depth and the smaller is the screening effect; additionally, there is a relatively greater penetration of fluid velocity toward the core. The implication of this dependence to the hydrodynamic ratio is examined below.

3.4.2. Relationship between the Saturation Penetration Depth, δ_e^*, and the Hydrodynamic Ratio, β. The hydrodynamic ratio, β, represents a measure of resistance offered by the aggregate. It is instructive to establish a quantitative relationship, if it exists, between β and the saturation penetration depth, δ_e^*, to improve our understanding of the fluid−aggregate interaction. It stands to reason to assume that an aggregate into which the fluid penetrates more easily will offer less resistance to flow. That is because the higher penetration of fluid into the aggregate will result in a smaller difference between the interior and exterior fluid velocities, leading to a lower resistance. The observation of a monotonic increase of β with d_f and a monotonic decrease of δ_e^* with d_f suggests a monotonic decrease of β with δ_e^*. This observation is in conformity with the reasoning that the degree of fluid penetration has a direct bearing on the hydrodynamic ratio. These observations permit us to assume a unique one-to-one relationship between β and δ_e^* as depicted in Figure 7b. To obtain this figure, we have coarse grained the data with respect

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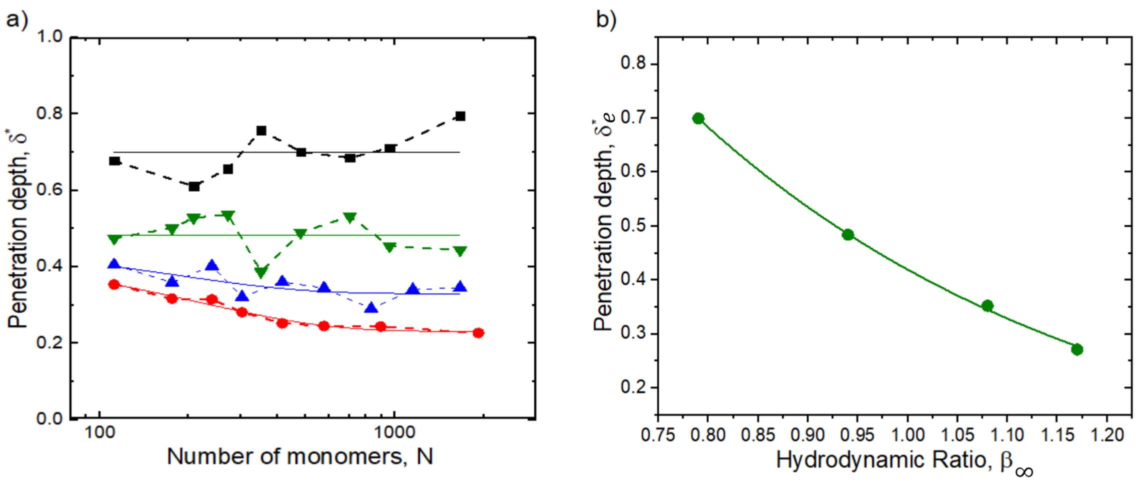

Figure 7. (a) N dependence of the penetration depth (δ^*) for tunable Monte Carlo clusters of different fractal dimensions along with the exponential fits (solid lines) (■, $d_f = 1.76$; green ▼, $d_f = 2$; blue ▲, $d_f = 2.25$; red ●, $d_f = 2.49$). (b) Variation of the asymptotic penetration depth δ_e^* with the hydrodynamic ratio β_∞ for the MC clusters. The continuous line is an exponential fit (−) to the data for tunable Monte Carlo clusters of different fractal dimensions.

Table 3. Penetration Depth, δ_e^*, versus d_f Obtained for Tunable Monte Carlo Clusters

fractal dimension (d_f)	saturation penetration depth (δ_e^*)
1.76	0.699 ± 0.08
1.99	0.484 ± 0.07
2.25	0.352 ± 0.05
2.49	0.271 ± 0.05

to nearby d_f values and considered the effect for 4 distinct d_f values. It may be noted that a 50% change in β leads to more than a 300% change in δ_e^*, thereby strongly pointing at a super-linear relationship. To capture this steep change, we attempt an exponential fit to the data within the range of β obtained. This results in the fitted function of the form:

$$\delta^* = (4.81 \pm 0.23) \exp\left(-\frac{\beta}{0.41 \pm 0.01}\right) \quad (9)$$

The above results indicate that the measure of the "cluster span" to fluid resistance changes only logarithmically with respect to the measure of its "transparency" to fluid penetration. This relationship offers a way to estimate the interior properties, such as the velocities at the core region, from the exterior measures such as the β. In catalysis, this could mean that the inner screened regions may contain active sites useful for chemical reactions.[14] Its implications in the context of aggregate mediated catalysis[13,14] or gas adsorption in which one would prefer deep penetration for more effective performance need to be explored in the future. For example, fractal models are increasingly being used in noninvasive methods of assessing flow obstructions in alveoli. It is worth exploring whether one can deploy resistance to flow as a surrogate parameter to estimate fluid penetration in these problems.

4. CONCLUSION

We have successfully applied, for the first time, a unified approach based on Stokesian dynamics formalism to obtain several hydrodynamic properties of rigid aggregates across a range of fractal dimensions between 1.76 and 3 and fractal types (mathematical, tunable Monte Carlo, and diffusion limited particle−cluster and cluster−cluster). This has resulted in the establishment of systematic correlations between the hydrodynamic ratio, β, the fractal dimension, d_f, and the cluster mass, N. These correlations are of considerable practical value to describe aggregate particle transport in the context of electrophoresis in liquid and gaseous systems, gravitational settling, and particle deposition in the lung for aerosols systems in the continuum regime. The study is required to be extended to noncontinuum regimes in the future to cover the entire particle size range.

The present study establishes a crucial relationship between hydrodynamic ratio β and the scaled penetration depth δ^* of the fluid into fractal clusters. The simulations indicate that the latter quantity tends to approach a nonzero steady value for large clusters as against the mean-field prediction of a monotonic decrease. This departure is plausible as the mean-field models invoke only local resistances of a given monomer and do not involve multiple, fluid−particle interactions as is done in Stokesian dynamics. The observation of a monotonic variation of the penetration depth on the fractal dimension and the hydrodynamic ratio implies a unique dependence on the degree of velocity penetration (an interior property) to the "effective" peripheral size offering resistance to flow (an external parameter). In other words, our study implies that a fractal aggregate having a higher hydrodynamic ratio implies lower fluid penetration as the underlying cause of its enhanced resistance. The finding that large changes occur in penetration depth for small changes in β implies a sublinear effect represented by a logarithmic form of fluid penetration in controlling the hydrodynamic mobility. These results have a potential bearing on the flow distribution within fractal catalysts, which in turn has a crucial implication for maximizing its functional efficacies. The hydrodynamic ratio could then act as a surrogate measure of the flow distribution, which may be useful for a quick assessment. This could be considered as a matter of future research for various applications in catalysis and gas adsorption phenomena. It is worth exploring whether external attributes such as the hydrodynamic ratio could serve as surrogate measures of internal mixing processes in these systems.

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ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.1c03180.

Aggregation generation parameters for fractals generated using tunable CCA; effect of the number of realizations on the average estimate of the hydrodynamic ratio; difference in values of $1/d_f$ and x with the increasing size of the cluster; comparison of the estimated hydrodynamic ratio with previous works; comparison of the estimated penetration depth with Davies' correlation (PDF)

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Notes

The authors declare no competing financial interest.

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NOMENCLATURE

d_f	fractal dimension
R_m	mobility radius
R_g	radius of gyration
δ	fluid penetration depth
δ^*	penetration depth scaled with R_g
β	hydrodynamic ratio
N	number of monomers
a	monomer radius
k_f	fractal prefactor
$\boldsymbol{r}_i$	position vector of i^{th} sphere
$\boldsymbol{r}_{cm}$	position vector of center of mass
R_c	outer radius
f	total force acting on aggregate
$\boldsymbol{u}^{\infty}$	fluid velocity
μ	fluid viscosity
κ	permeability
p	pressure
r	radial position from the center of mass of the aggregate
C	initial monomer number density on the lattice
L	length of cubic lattice
γ	diffusivity exponent
$\mathbf{U}$	$6N \times 1$ vector representing translational and rotational velocity components of each monomer particle
$\mathbf{F}^P$	$6N \times 1$ vector representing force and torque components on the monomer particle
$\mathbf{R}_{FU}$	$6N \times 6N$ resistance matrix
M^{∞}	$6N \times 6N$ mobility matrix
$\boldsymbol{\Sigma}$	$6N \times 6$ rigid body matrix
V	characteristic velocity
J	Oseen's tensor
$\mathbf{U}^{\infty}$	$6N \times 1$ vector representing translational and rotational velocity components of fluid velocity
k_m	mobility prefactor
x	mobility exponent
β_{∞}	asymptotic value of hydrodynamic ratio
A	constant
N_0	cluster size beyond which the constant hydrodynamic ratio is reached
r^*	radial distance from the center of mass scaled by the radius of gyration
u^*	dimensionless velocity of fluid scaled by asymptotic fluid velocity
δ^*	penetration depth scaled by the radius of gyration
δ_e^*	asymptotic value of dimensionless penetration depth

REFERENCES

(1) McMurry, P. H. A Review of Atmospheric Aerosol Measurements. *Atmos. Environ.* **2000**, *34*, 1959.

(2) Venkataraman, C.; Reddy, C. K.; Josson, S.; Reddy, M. S. Aerosol Size and Chemical Characteristics at Mumbai, India, during the INDOEX-IFP (1999). *Atmos. Environ.* **2002**, *36*, 1979.

(3) Mallik, A. K.; Mukherjee, S.; Panchagnula, M. V. An Experimental Study of Respiratory Aerosol Transport in Phantom Lung Bronchioles. *Phys. Fluids* **2020**, *32*, 111903.

(4) Gurazada, M.; Karambelas, A.; Chhatre, A. *Impact of Emissions from Coal-Fired Power Plants on Health in India*; https://www.isb.edu/en/research-thought-leadership/research-centres-institutes/bharti-institute-of-public-policy/Research/Reports/reportitem3.html.

(5) Shein, E. V.; Devin, B. A. Current Problems in the Study of Colloidal Transport in Soil. *Eurasian Soil Science* **2007**, *40*, 399.

(6) Xu, S.; Gao, B.; Saiers, J. E. Straining of Colloidal Particles in Saturated Porous Media. *Water Resour. Res.* **2006**, *42*, 1.

(7) McGechan, M. B.; Lewis, D. R. Transport of Particulate and Colloid-Sorbed Contaminants through Soil, Part 1: General Principles. *Biosystems Engineering* **2002**, *83*, 255.

(8) Jacobson, M. Z. Control of Fossil-Fuel Particulate Black Carbon and Organic Matter, Possibly the Most Effective Method of Slowing Global Warming. *Journal of Geophysical Research Atmospheres* **2002**, *107*, ACH 16-1.

(9) Mittal, R.; Ni, R.; Seo, J.-H. The Flow Physics of COVID-19. *Journal of fluid Mechanics* **2020**, *894*, 894.

(10) Mackay, D. M.; Roberts, P. V.; Cherry, J. A. Transport of Organic Contaminants in Groundwater. *Environ. Sci. Technol.* **1985**, *19* (5), 384−392.

(11) Backhus, D. A.; Ryan, J. N.; Groher, D. M.; MacFarlane, J. K.; Gschwend, P. M. Sampling Colloids and Colloid-Associated Contaminants in Ground Water. *Groundwater* **1993**, *31*, 466.
(12) Lieberman, H.; Rieger, M.; Banker, G. S. *Pharmaceutical Dosage Forms: Disperse Systems*; CRC Press, 2020.
(13) Sheintuch, M. Reaction Engineering Principles of Processes Catalyzed by Fractal Solids. *Catalysis Reviews - Science and Engineering* **2001**, *43*, 233.
(14) Bukowski, B. C.; Keil, F. J.; Ravikovitch, P. I.; Sastre, G.; Snurr, R. Q.; Coppens, M. O. Connecting Theory and Simulation with Experiment for the Study of Diffusion in Nanoporous Solids. *Adsorption* **2021**, *27*, 683.
(15) NEA. *State-of-the-Art Report (SOAR) on Nuclear Aerosols*; 2009; https://www.oecd-nea.org/jcms/pl_18750/state-of-the-art-report-soar-on-nuclear-aerosols.
(16) Dickinson, S.; Auvinen, A.; Ammar, Y.; Bosland, L.; Clément, B.; Funke, F.; Glowa, G.; Kärkelä, T.; Powers, D. A.; Tietze, S.; Weber, G.; Zhang, S. Experimental and Modelling Studies of Iodine Oxide Formation and Aerosol Behaviour Relevant to Nuclear Reactor Accidents. *Annals of Nuclear Energy* **2014**, *74*, 200.
(17) Samson, R. J.; Mulholland, G. W.; Gentry, J. W. Structural Analysis of Soot Agglomerates. *Langmuir* **1987**, *3*, 272.
(18) Tozawa, H.; Kato, Y.; Sorimachi, K.; Nakanishi, T. Agglomeration and flotation of alumina clusters in molten steel. *ISIJ International* **1999**, *39*, 426.
(19) Jing, D.; Sun, L.; Jin, J.; Thangamuthu, M.; Tang, J. Magneto-Optical Transmission in Magnetic Nanoparticle Suspensions for Different Optical Applications: A Review. *J. Phys. D: Appl. Phys.* **2021**, *54*, 013001.
(20) Jing, D.; Hu, S.; Zhang, Y.; Luo, J. A Modified Diffusion-Limited Cluster Aggregation Model for Accurate Prediction of the Coagulation and Fragmentation Process in Nanoparticle Suspension. *J. Phys. D: Appl. Phys.* **2019**, *52*, 455305.
(21) Bonaccorsi, L.; Calandra, P.; Amenitsch, H.; Proverbio, E.; Lombardo, D. Growth of Fractal Aggregates during Template Directed SAPO-34 Zeolite Formation. *Microporous Mesoporous Mater.* **2013**, *167*, 3.
(22) Yu, M.; Lin, J. Taylor-Expansion Moment Method for Agglomerate Coagulation Due to Brownian Motion in the Entire Size Regime. *J. Aerosol Sci.* **2009**, *40* (6), 549.
(23) Gonfiotti, B.; Paci, S. Standalone Containment Analysis of Four Phébus Tests with the ASTEC and the MELCOR Codes. *Science and Technology of Nuclear Installations* **2019**, *2019* (3), No. 6021346.
(24) Zhang, B.; Sheng, W.; Xu, M.; Zhang, Y.; Lei, J. Y.; Ma, H. C. The Influence of Aerosol Density and Shape Factor on the Assessment of Internal Exposure to 239Pu. *Health Physics* **2019**, *117*, 353.
(25) Kotrappa, P.; Sundararajan, A. R.; Bhanti, D. P.; Menon, V. B. Dynamic Shape Factors for Puo2 Aerosols Useful in Autoradiographic Particle Size Analysis. *Health physics* **1975**, *29* (5), 701−704.
(26) Forrest, S. R.; Witten, T. A. Long-Range Correlations in Smoke-Particle Aggregates. *Journal of Physics A: General Physics* **1979**, *12*, L109.
(27) Adler, P. M. Hydrodynamic Properties of Fractal Flocs. *Faraday Discuss. Chem. Soc.* **1987**, *83*, 145−152.
(28) Degiorgio, V. Mandelbrot, BB The Fractal Geometry Of Nature. *Scientia, Rivista di Scienza* **1984**, *78* (119), 27.
(29) Nagel, S. R. Experimental Soft-Matter Science. *Rev. Mod. Phys.* **2017**, *89*, 025002.
(30) DeCarlo, P. F.; Slowik, J. G.; Worsnop, D. R.; Davidovits, P.; Jimenez, J. L. Particle Morphology and Density Characterization by Combined Mobility and Aerodynamic Diameter Measurements. Part 1: Theory. *Aerosol Sci. Technol.* **2004**, *38*, 1185.
(31) Nguyen, P. H. *Numerical Simulations of the Lattice Boltzmann Method for Determination of Hydrodynamic Properties of Fractal Aggregates*. Ph.D. Thesis, University of Geneva, 2007.
(32) Kirkwood, J. G.; Riseman, J. The Intrinsic Viscosities and Diffusion Constants of Flexible Macromolecules in Solution. *J. Chem. Phys.* **1948**, *16*, 565.
(33) Brinkman, H. C. A Calculation of the Viscosity and the Sedimentation Constant for Solutions of Large Chain Molecules Taking into Account the Hampered Flow of the Solvent through These Molecules. *Physica* **1947**, *13*, 447.
(34) Beavers, G. S.; Joseph, D. D. Boundary Conditions at a Naturally Permeable Wall. *J. Fluid Mech.* **1967**, *30*, 197.
(35) Veerapaneni, S.; Wiesner, M. R. Hydrodynamics of Fractal Aggregates with Radially Varying Permeability. *J. Colloid Interface Sci.* **1996**, *177*, 45.
(36) Kim, A. S.; Yuan, R. Hydrodynamics of an Ideal Aggregate with Quadratically Increasing Permeability. *J. Colloid Interface Sci.* **2005**, *285*, 627.
(37) Wiltzius, P. Hydrodynamic Behavior of Fractal Aggregates. *Phys. Rev. Lett.* **1987**, *58*, 710.
(38) De Gennes, P. G. *Scaling Concepts in Polymer Physics*; Cornell University Press: Ithaca, NY, 1979.
(39) Van Saarloos, W. On the Hydrodynamic Radius of Fractal Aggregates. *Physica A: Statistical Mechanics and its Applications* **1987**, *147*, 280.
(40) Chen, Z. Y.; Weakliem, P. C.; Meakin, P. Hydrodynamic Radii of Diffusion-Limited Aggregates and Bond-Percolation Clusters. *J. Chem. Phys.* **1988**, *89*, 5887.
(41) Warren, P. B. Hydrodynamics of Fractal Aggregates. *Il Nuovo Cimento D* **1994**, *16*, 1231.
(42) Gmachowski, L. Hydrodynamics of Aggregated Media. *J. Colloid Interface Sci.* **1996**, *178*, 80.
(43) Fellay, L. S.; Twist, C.; Vanni, M. Motion of Rigid Aggregates under Different Flow Conditions. *Acta Mechanica* **2013**, *224*, 2225.
(44) Chopard, B.; Nguyen, H.; Stoll, S. A Lattice Boltzmann Study of the Hydrodynamic Properties of 3D Fractal Aggregates. *Mathematics and Computers in Simulation* **2006**, *72*, 103.
(45) Harshe, Y. M.; Ehrl, L.; Lattuada, M. Hydrodynamic Properties of Rigid Fractal Aggregates of Arbitrary Morphology. *J. Colloid Interface Sci.* **2010**, *352*, 87.
(46) Vanni, M. Creeping Flow over Spherical Permeable Aggregates. *Chem. Eng. Sci.* **2000**, *55*, 685.
(47) Witten, T. A.; Sander, L. M. Diffusion-Limited Aggregation, a Kinetic Critical Phenomenon. *Phys. Rev. Lett.* **1981**, *47*, 1400.
(48) Woignier, T.; Rangon, L.; Clostre, F.; Mottes, C.; Cattan, P.; Primera, J.; Jannoyer, M. Physical Limitation of Pesticides (Chlordecone) Decontamination in Volcanic Soils: Fractal Approach and Numerical Simulation. *Environmental Science and Pollution Research* **2020**, *27*, 40980.
(49) Heinson, W. R.; Liu, P.; Chakrabarty, R. K. Fractal Scaling of Coated Soot Aggregates. *Aerosol Sci. Technol.* **2017**, *51*, 12.
(50) Li, C.; Xiong, H. 3D Simulation of the Cluster-Cluster Aggregation Model. *Comput. Phys. Commun.* **2014**, *185*, 3424.
(51) Filippov, A. V.; Zurita, M.; Rosner, D. E. Fractal-like Aggregates: Relation between Morphology and Physical Properties. *J. Colloid Interface Sci.* **2000**, *229*, 261.
(52) Moteki, N. Capabilities and Limitations of the Single-Particle Extinction and Scattering Method for Estimating the Complex Refractive Index and Size-Distribution of Spherical and Non-Spherical Submicron Particles. *Journal of Quantitative Spectroscopy and Radiative Transfer* **2020**, *243*, 106811.
(53) Brady, J. F.; Bossis, G. Stokesian Dynamics. *Annu. Rev. Fluid Mech.* **1988**, *20*, 111.
(54) Townsend, A. K. *The Mechanics of Suspensions*. Ph.D. Thesis, University College London, 2017.
(55) Durlofsky, L.; Brady, J. F. Analysis of the Brinkman Equation as a Model for Flow in Porous Media. *Phys. Fluids* **1987**, *30*, 3329.
(56) Swan, J. W.; Brady, J. F.; Moore, R. S.; Dooling, L.; Hoh, N.; Choi, J.; Zia, R. Modeling Hydrodynamic Self-Propulsion with Stokesian Dynamics. Or Teaching Stokesian Dynamics to Swim. *Phys. Fluids* **2011**, *23*, 071901.
(57) Kutteh, R. Rigid Body Dynamics Approach to Stokesian Dynamics Simulations of Nonspherical Particles. *J. Chem. Phys.* **2010**, *132*, 174107.

https://doi.org/10.1021/acs.langmuir.1c03180
Langmuir 2022, 38, 3422−3433

(58) Ichiki, K.; Brady, J. F. Many-Body Effects and Matrix Inversion in Low-Reynolds-Number Hydrodynamics. *Phys. Fluids* **2001**, *13*, 350.
(59) Kim, S.; Karrila, S. J. *Microhydrodynamics: Principles And Selected Applications*; Dover Publications Incorporated, 2005.
(60) Bossis, G.; Meunier, A.; Brady, J. F. Hydrodynamic Stress on Fractal Aggregates of Spheres. *J. Chem. Phys.* **1991**, *94* (7), 5064−5070.
(61) Swan, J. W.; Brady, J. F.; Moore, R. S. Modeling Hydrodynamic Self-Propulsion with Stokesian Dynamics. Or Teaching Stokesian Dynamics to Swim. *Phys. Fluids* **2011**, *23* (7), 071901.
(62) Sorensen, C. M. The Mobility of Fractal Aggregates: A Review. *Aerosol Sci. Technol.* **2011**, *45*, 765.
(63) Seto, R.; Botet, R.; Briesen, H. Hydrodynamic Stress on Small Colloidal Aggregates in Shear Flow Using Stokesian Dynamics. *Phys. Rev. E* **2011**, *84* (4), 041405.
(64) Dastanpour, R.; Rogak, S. N. The Effect of Primary Particle Polydispersity on the Morphology and Mobility Diameter of the Fractal Agglomerates in Different Flow Regimes. *J. Aerosol Sci.* **2016**, *94*, 22.
(65) Pusey, P. N.; Rarity, J. G. Measurement of the Hydrodynamic Fractal Dimension of Aggregating Polystyrene Spheres. *Mol. Phys.* **1987**, *62* (2), 411−418.
(66) Wang, G. M.; Sorensen, C. M. Diffusive Mobility of Fractal Aggregates over the Entire Knudsen Number Range. *Phys. Rev. E* **1999**, *60* (3), 3036−3044.
(67) Debye, P.; Bueche, A. M. Intrinsic Viscosity, Diffusion, and Sedimentation Rate of Polymers in Solution. *J. Chem. Phys.* **1948**, *16*, 573.
(68) Hwang, C. O.; Given, J. A.; Mascagni, M. On the Rapid Estimation of Permeability for Porous Media Using Brownian Motion Paths. *Phys. Fluids* **2000**, *12*, 1699.
(69) Davies, C. N. The Separation of Airborne Dust and Particles. *Proceedings of the Institution of Mechanical Engineers, Part B: Management and engineering manufacture* **1953**, *167*, 185−213.
(70) Kim, S.; Russel, W. B. Modelling of Porous Media by Renormalization of the Stokes Equations. *J. Fluid Mech.* **1985**, *154*, 269.

https://doi.org/10.1021/acs.langmuir.1c03180
Langmuir 2022, 38, 3422−3433